

EXPERIMENT-6

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Date: 19/08/2026
Subject Code: 24CSH-337

AIM: To implement VLAN (Virtual Local Area Network) and VLAN Trunking protocols in Cisco Packet Tracer and to understand how VLANs provide logical network segmentation and how trunk links carry traffic belonging to multiple VLANs between switches.

OBJECTIVES:

- To understand the concept of VLAN and its importance in computer networks.
- To understand how VLANs logically divide a physical network into multiple broadcast domains.
- To differentiate between Access Ports and Trunk Ports.
- To configure VLANs on Cisco switches using Cisco IOS commands.
- To assign switch ports to specific VLANs.
- To understand the working of VLAN trunking.
- To understand VLAN tagging using the IEEE 802.1Q standard.
- To configure a trunk link between two Cisco switches.
- To verify VLAN configuration using appropriate Cisco IOS commands.
- To test connectivity between devices belonging to the same and different VLANs.

SOFTWARE REQUIREMENTS:

(a) Software:

- Cisco Packet Tracer
- Windows Operating System

(b) Hardware (Virtual):

- PC-PT

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- Cisco 2960-24TT Switch
- Copper Straight-Through Cable

THEORY: (i) Layer 2 Switch:

A Layer 2 switch is a network device that primarily operates at the Data Link Layer (Layer 2) of the OSI model.

It uses MAC addresses to forward Ethernet frames.

For example:

PC1 → Switch → PC2

The switch receives an Ethernet frame, examines the destination MAC address and forwards the frame through the appropriate port.

The switch maintains a MAC Address Table to determine which device is connected to which port.

Eg:-

MAC Address	Port
PC1 MAC	Fa0/1
PC2 MAC	Fa0/2
PC3 MAC	Fa0/3

If a frame is destined for PC2, the switch identifies that PC2 is connected to Fa0/2 and forwards the frame through that port.

The Cisco 2960 switch commonly used in Packet Tracer is primarily a Layer 2 switch.

(ii) Layer 3 Switch:

A Layer 3 switch can perform both:

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- Layer 2 switching
- Layer 3 routing

It can understand:

- MAC addresses
- IP addresses

A Layer 2 switch cannot normally route traffic between different VLANs. A Layer 3 switch can perform Inter-VLAN Routing.

Feature	Layer 2 Switch	Layer 3 Switch
Main OSI Layer	Layer 2	Layer 2 + Layer 3
Uses MAC Address	Yes	Yes
Uses IP Address for Routing	No	Yes
Switching	Yes	Yes
Routing	No	Yes
Inter-VLAN Routing	Not by itself	Yes
Example	Cisco 2960	Cisco 3560

(iii) What is VLAN?

VLAN stands for Virtual Local Area Network.

A VLAN allows a single physical switch to be logically divided into multiple separate networks.

Without VLANs, all computers connected to the same switch normally belong to the same Layer 2 broadcast domain.

For example, an organization may have:

- HR Department
- Sales Department

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Instead of using two physically separate switches, one physical switch can be logically divided using VLANs.

Therefore, VLANs create multiple logical networks within the same physical switch.

(iv) Working of VLAN:

Suppose:

VLAN 10 → HR

VLAN 20 → Sales

The network can be represented as:

- PC1 → VLAN 10
- PC2 → VLAN 10
- PC3 → VLAN 20
- PC4 → VLAN 20

PC1 and PC2 belong to VLAN 10, while PC3 and PC4 belong to VLAN 20.

Therefore:

PC1 → PC2 → Communication possible

PC3 → PC4 → Communication possible

PC1 → PC3 → Communication not possible without Layer 3 routing.

VLAN 10 and VLAN 20 are treated as separate Layer 2 networks.

(v) Why VLANs are needed?

Problem in Normal LAN	How VLAN Solves It	Why It Is Needed
All devices are in the same broadcast domain	VLAN divides one physical LAN into multiple logical networks	Reduces unnecessary broadcast traffic

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Difficult to separate departments	Different departments can use different VLANs	Provides logical organization
Reduced security	Devices in different VLANs are logically separated	Improves network segmentation
Broadcast traffic reaches unnecessary devices	Broadcasts are limited to the same VLAN	Improves network performance
Physical location determines grouping	VLAN allows logical grouping regardless of location	Easier network management
Moving a user may require physical changes	Switch port can be assigned to the required VLAN	Simplifies administration
Large LAN becomes difficult to manage	Large LAN can be divided into smaller VLANs	Easier management
Departments need separate networks	Each department can have its own VLAN	Provides logical separation
Broadcast storms can affect a large LAN	VLANs create separate broadcast domains	Limits the impact
One switch needs to support multiple networks	Multiple VLANs can exist on one physical switch	Saves hardware and infrastructure cost

(vi) Example of VLAN in an Organization:

Consider a college network containing:

- Students
- Faculty
- Administration
- Accounts

Instead of placing everyone in one large network, VLANs can be created:

VLAN 10 → Students

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VLAN 20 → Faculty

VLAN 30 → Administration

VLAN 40 → Accounts

Even though all departments may use the same physical switch infrastructure, they remain logically separated.

(vii) Access Ports:

An access port is a switch port that normally belongs to one VLAN.

For example:

PC1 → Fa0/1 → Switch

If Fa0/1 is configured for VLAN 10, PC1 becomes a member of VLAN 10.

Similarly:

PC2 → Fa0/2 → Switch

If Fa0/2 is configured for VLAN 20, PC2 becomes a member of VLAN 20.

Access ports are generally used to connect:

- PCs
 - Printers
 - Servers
 - End-user devices
 - IP phones in appropriate configurations
-

(viii) Trunk Ports:

Suppose we have two switches and both switches contain:

VLAN 10 → HR

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VLAN 20 → Sales

We need to connect the two switches.

Using separate physical cables for every VLAN would be inefficient.

Instead, a Trunk Port is used.

A trunk port is a switch port that can carry traffic belonging to multiple VLANs over a single physical link.

Therefore, one physical cable can carry traffic for VLAN 10, VLAN 20, VLAN 30, etc.

(ix) Access Port vs Trunk Port:

Access Port	Trunk Port
Normally carries one VLAN	Carries multiple VLANs
Usually connects end devices	Usually connects network devices
PC → Switch	Switch → Switch
Traffic toward end device is normally untagged	VLAN information is carried across the trunk
Example: Fa0/1 assigned to VLAN 10	Example: Gi0/1 carrying VLAN 10,20,30

(x) VLAN Tagging:

When traffic belonging to multiple VLANs travels through a trunk link, the receiving switch needs to identify which VLAN each frame belongs to.

This is achieved using VLAN tagging.

The most important standard used for Ethernet VLAN trunking is:

IEEE 802.1Q

A VLAN tag contains information that allows the receiving switch to identify the VLAN associated with a frame.

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For example:

Frame → VLAN ID = 10

The receiving switch sees the VLAN ID and understands that the frame belongs to VLAN 10.

(xi) VLAN Trunking:

VLAN trunking is the process of carrying traffic from multiple VLANs over a single physical network link between network devices.

For example:

SW1 Gi0/1 ⇔ Gi0/1 SW2

The link can carry:

- VLAN 10 traffic
- VLAN 20 traffic

using the trunking mechanism.

(xii) VLAN Trunking ≠ VTP:

There are three different concepts:

1. VLAN Trunking

Carries traffic belonging to multiple VLANs over one physical link.

2. IEEE 802.1Q

A standard used to tag VLAN traffic on an Ethernet trunk.

3. VTP — VLAN Trunking Protocol

VTP is used to distribute/manage VLAN configuration information among Cisco switches in a VTP domain.

Therefore, VTP is not the protocol that carries multiple VLANs over a trunk.

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(xiii) Types of Trunking Protocols:

Protocol	Type	Status/Use
IEEE 802.1Q	Open IEEE standard	Common/current Ethernet VLAN trunking method
ISL	Cisco proprietary	Legacy/obsolete in modern networks

For this experiment, IEEE 802.1Q trunking is used.

STEPS/IMPLEMENTATION:

Part I – Creating the Network Topology:

1. Open Cisco Packet Tracer.
2. Place two Cisco 2960-24TT switches in the workspace.
3. Place four PC-PT devices.
4. Connect the devices using Copper Straight-Through cables as follows:

Device	Connected To	Port
PC1	Switch1	Fa0/1
PC2	Switch1	Fa0/2
PC3	Switch2	Fa0/1
PC4	Switch2	Fa0/2
Switch1	Switch2	Gi0/1

5. The connection between Switch1 and Switch2 will be configured as the TRUNK link.

The topology should contain two logical departments:

- HR → VLAN 10
- SALES → VLAN 20

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Part II – Assigning IP Addresses:

Configure the IP addresses from PC → Desktop → IP Configuration.

Device	IP Address	Subnet Mask	VLAN
PC1	192.168.1.1	255.255.255.0	VLAN 10 – HR
PC2	192.168.2.1	255.255.255.0	VLAN 20 – SALES
PC3	192.168.1.2	255.255.255.0	VLAN 10 – HR
PC4	192.168.2.2	255.255.255.0	VLAN 20 – SALES

PC1 and PC3 belong to the HR VLAN (VLAN 10).

PC2 and PC4 belong to the SALES VLAN (VLAN 20).

Part III- Configure VLAN 10 on Switch1:

1. Open Switch1 → CLI.
2. Enter the following commands:

enable

configure terminal

vlan 10

name HR

exit

VLAN 10 is now created and named HR.

Part IV – Configure VLAN 20 on Switch1:

Enter: vlan 20

name SALES

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exit

Switch1 now contains:

- VLAN 10 → HR
- VLAN 20 → SALES

Part V – Assign Fa0/1 to VLAN 10:

Configure the port connected to PC1:

```
interface fa0/1
```

```
switchport mode access
```

```
switchport access vlan 10
```

```
exit
```

Therefore:

PC1 → Fa0/1 → VLAN 10 (HR)

Part VI - Assign Fa0/2 to VLAN 20:

Configure the port connected to PC2:

```
interface fa0/2
```

```
switchport mode access
```

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switchport access vlan 20

exit

Therefore:

PC2 → Fa0/2 → VLAN 20 (SALES)

Part VII - Configure VLANs on Switch2:

Open Switch2 → CLI and enter:

enable

configure terminal

vlan 10

name HR

exit

vlan 20

name SALES

exit

Switch2 now also has:

- VLAN 10 → HR

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- VLAN 20 → SALES

Step VIII - Assign Fa0/1 to VLAN 10 on Switch2:

Configure the port connected to PC3:

```
interface fa0/1
```

```
switchport mode access
```

```
switchport access vlan 10
```

```
exit
```

Therefore:

PC3 → Fa0/1 → VLAN 10 (HR)

Step IX - Assign Fa0/2 to VLAN 20 on Switch2:

Configure the port connected to PC4:

```
interface fa0/2
```

```
switchport mode access
```

```
switchport access vlan 20
```

```
exit
```

Therefore:

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PC4 → Fa0/2 → VLAN 20 (SALES)

Step X - Configure Trunk Port on Switch1:

The connection between the switches is:

Switch1 Gi0/1 ===== Gi0/1 Switch2

On Switch1, enter:

enable

configure terminal

interface gigabitEthernet 0/1

switchport mode trunk

exit

The Gi0/1 port is now configured as a trunk.

Step XI - Configure Trunk Port on Switch2:

On Switch2, enter:

enable

configure terminal

interface gigabitEthernet 0/1

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switchport mode trunk

exit

Now the link between Switch1 and Switch2 is a trunk link capable of carrying traffic from both VLAN 10 and VLAN 20.

Step XII - Verify VLAN Configuration:

On both switches, enter:

show vlan brief

Verify that:

- Fa0/1 → VLAN 10 (HR)
- Fa0/2 → VLAN 20 (SALES)

Step XIII - Verify Trunk Configuration:

Enter:

show interfaces trunk

The output should show GigabitEthernet0/1 as a trunk port.

The trunk allows VLAN 10 and VLAN 20 traffic to travel between the two switches over the same physical connection.

Step XIV – Verify MAC Address Table:

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Enter:

```
show mac address-table
```

This command displays the MAC addresses learned by the switch and the corresponding ports/VLANs.

Step XV - Test Connectivity:

Test 1: PC1 → PC3

Open PC1 → Desktop → Command Prompt and enter:

```
ping 192.168.1.2
```

Expected Result:

Successful

PC1 and PC3 both belong to VLAN 10 (HR).

Test 2: PC2 → PC4

On PC2, enter:

```
ping 192.168.2.2
```

Expected Result:

Successful

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PC2 and PC4 both belong to VLAN 20 (SALES).

Test 3: PC1 → PC4

On PC1, enter:

ping 192.168.2.2

Expected Result:

Unsuccessful

Reason:

PC1 → VLAN 10 (HR)

PC4 → VLAN 20 (SALES)

Since the experiment does not configure a router or Layer 3 switch for Inter-VLAN Routing, devices from different VLANs cannot communicate directly.

Step XVI - Observe VLAN Trunking:

When PC1 sends a frame to PC3:

PC1 → Switch1 → TRUNK → Switch2 → PC3

The frame belongs to VLAN 10.

Similarly, traffic between PC2 and PC4 travels through the same trunk link but belongs to VLAN 20.

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Thus, a single physical link between the switches carries traffic for multiple VLANs.

PROCEDURE:

1. Network Topology

- Open Cisco Packet Tracer.
- Place two Cisco 2960 switches and four PCs.
- Connect PC1 and PC2 to Switch1.
- Connect PC3 and PC4 to Switch2.
- Connect Switch1 and Switch2 using GigabitEthernet ports.

2. IP Configuration

- Configure PC1 as 192.168.1.1/24.
- Configure PC3 as 192.168.1.2/24.
- Configure PC2 as 192.168.2.1/24.
- Configure PC4 as 192.168.2.2/24.

3. VLAN Configuration

- Create VLAN 10 – HR.
- Create VLAN 20 – SALES.
- Assign Fa0/1 to VLAN 10.
- Assign Fa0/2 to VLAN 20.
- Repeat the configuration on both switches.

4. Trunk Configuration

- Configure Gi0/1 of Switch1 as a trunk.
- Configure Gi0/1 of Switch2 as a trunk.
- Verify that the trunk is operational.

5. Verification

Use:

show vlan brief

to verify VLANs.

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Use:

show interfaces trunk

to verify the trunk.

Use:

show mac address-table

to verify MAC address learning.

6. Connectivity Testing

Use the ping command to test communication between devices in the same VLAN and different VLANs.

INPUT/OUTPUT DETAILS:

(a) Devices Used

- PC-PT × 4
- Cisco 2960-24TT Switch × 2
- Copper Straight-Through Cable

(b) VLANs Used

- VLAN 10 → HR
- VLAN 20 → SALES

(c) IP Networks Used

- 192.168.1.0/24 → HR
- 192.168.2.0/24 → SALES

(d) Important Cisco IOS Commands

enable

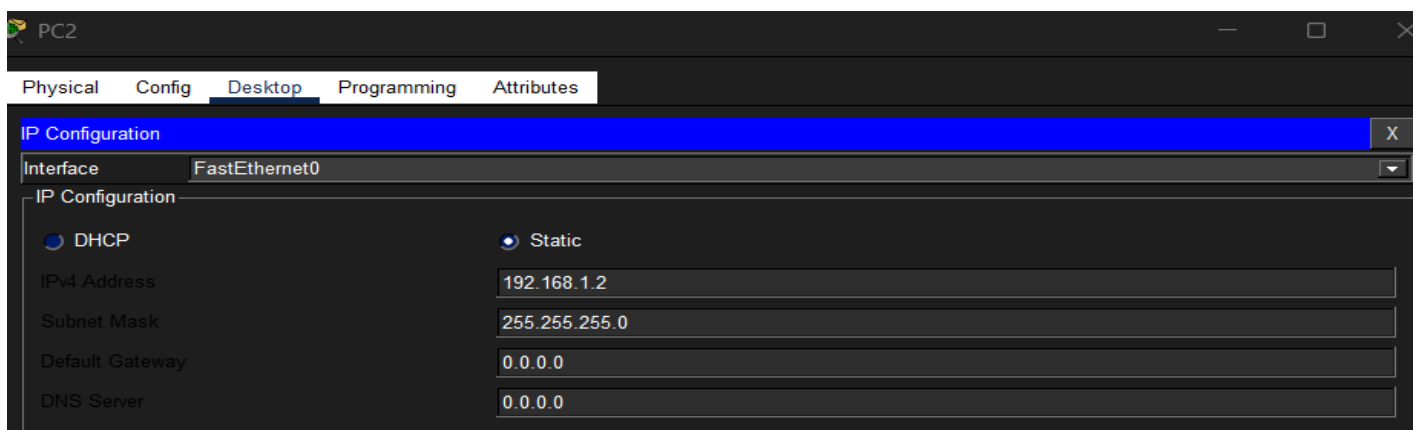
configure terminal

vlan 10

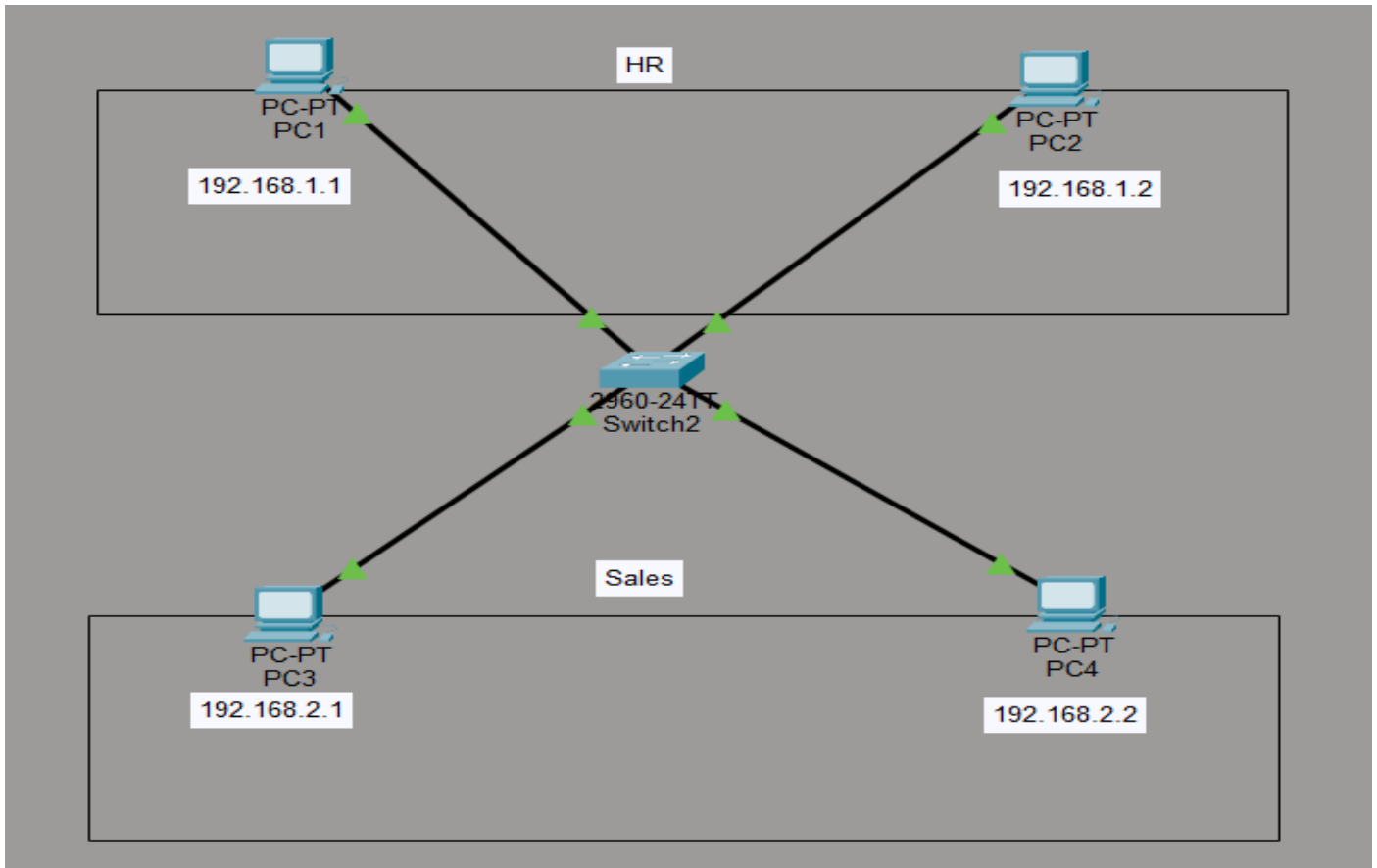
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```
name HR
vlan 20
name SALES
interface fa0/1
switchport mode access
switchport access vlan 10
interface fa0/2
switchport mode access
switchport access vlan 20
interface gigabitEthernet 0/1
switchport mode trunk
show vlan brief
show interfaces trunk
show mac address-table
```

DEMONSTRATION/OUTPUT:



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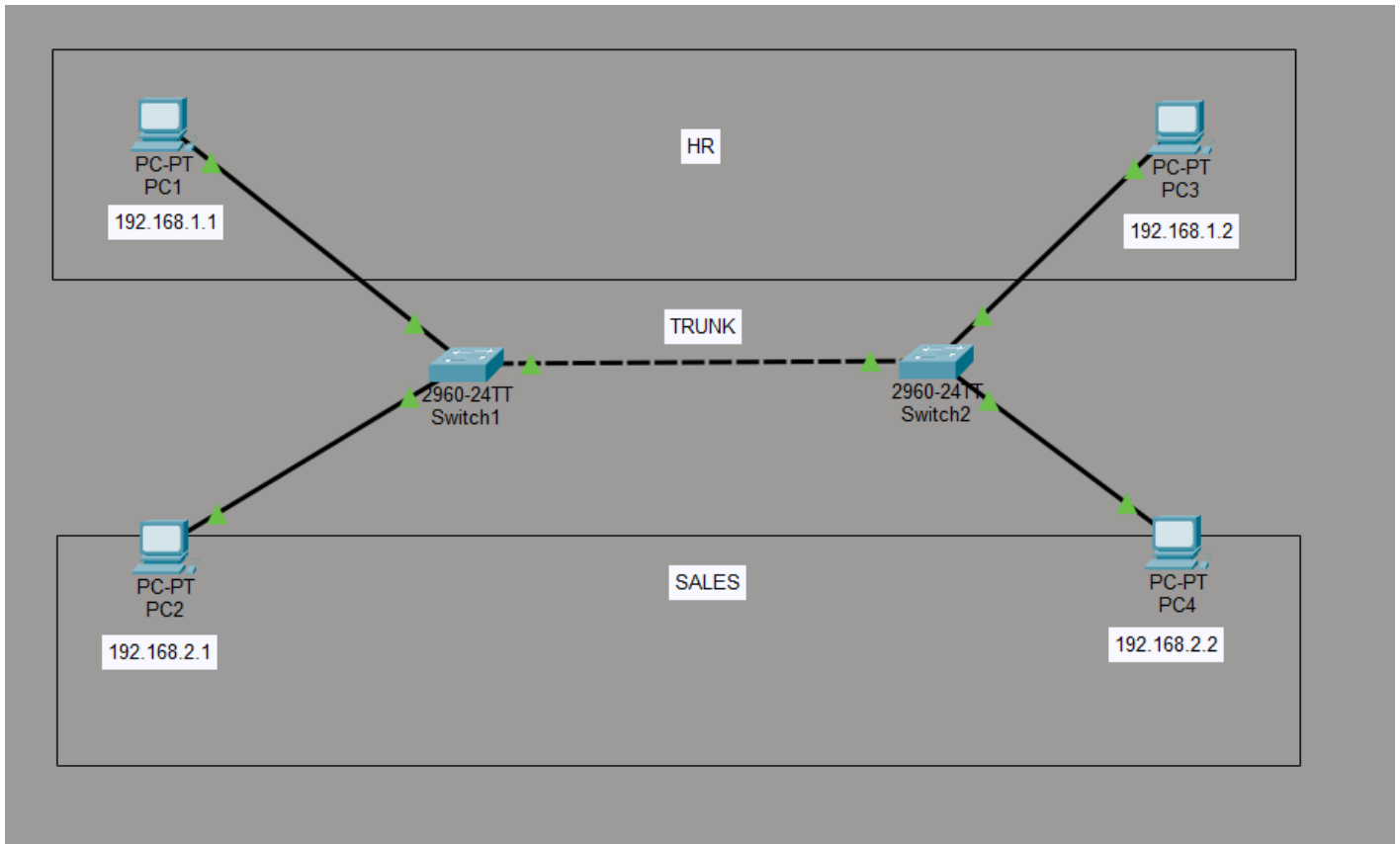
```

cisco WS-C2960-24TT-L (PowerPC405) processor (revision B0) with 65536K bytes of memory.
Processor board ID FOC1010X104
Last reset from power-on
1 Virtual Ethernet interface
24 FastEthernet interfaces
2 Gigabit Ethernet interfaces
The password-recovery mechanism is enabled.
64K bytes of flash-simulated non-volatile configuration memory.
Base ethernet MAC Address      : 00:60:70:75:75:C3
Motherboard assembly number    : 73-10390-03
Power supply part number       : 341-0097-02
Motherboard serial number      : FOC10093R12
Power supply serial number     : AZS1007032H
Model revision number          : B0
Motherboard revision number    : B0
Model number                   : WS-C2960-24TT-L
System serial number           : FOC1010X104
Top Assembly Part Number       : 800-27221-02
Top Assembly Revision Number   : A0
Version ID                     : V02
CLEI Code Number               : COM3L00BRA
Hardware Board Revision Number : 0x01
  
```

Switch Ports	Model	SW Version	SW Image
* 1 26	WS-C2960-24TT-L	15.0(2)SE4	C2960-LANBASEK9-M

Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version 15.0(2)SE4, RELEASE SOFTWARE (fc1)
 Technical Support: <http://www.cisco.com/techsupport>
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Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC1	PC3	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC2	PC4	ICMP		0.000	N	1	(edit)	(delete)

```

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
  
```

RESULT: The experiment was successfully completed in Cisco Packet Tracer. VLAN 10 (HR) and VLAN 20 (SALES) were created and configured on both switches. Access ports were

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assigned to the respective VLANs, and the link between the two switches was configured as a trunk. Connectivity between devices in the same VLAN was successfully verified, while communication between different VLANs was not possible without Inter-VLAN Routing.

LEARNING OUTCOMES:

After completing this experiment:

- I understood the concept of VLANs and their importance in logically segmenting a network.
- I learned how to create VLANs and assign switch ports to specific VLANs using Cisco IOS commands.
- I understood the difference between access ports and trunk ports and their applications.
- I learned how VLAN trunking allows multiple VLANs to communicate across a single physical link using VLAN tagging.
- I developed practical skills in verifying VLANs, trunk configuration, and network connectivity using Cisco Packet Tracer commands and the Ping utility.